# CompositionMeasure for RenyiDivergence

Michael Shoemate (@Shoeboxam)

This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of the implementation of `CompositionMeasure` for `RenyiDivergence` in `mod.rs` at commit f5bb719 (outdated[1]).

## 1 Hoare Triple

### Precondition

#### Compiler-verified

Types matching pseudocode.

#### Caller-verified

None

### Pseudocode

```
1  class CompositionMeasure(RenyiDivergence):
2      def composability(  #
3          self, adaptivity: Adaptivity
4      ) -> Composability:
5          return Composability.Concurrent
6
7      def compose(self, d_mids: Vec[Self_Distance]) -> Self_Distance:
8          def curve(alpha: float) -> float: #
9              epsilons = [d_mid(alpha) for d_mid in d_mids]
10
11             d_out = 0.0
12             for d_mid in epsilons:
13                 d_out = d_out.inf_add(d_mid)
14             return d_out
15
16         return Function.new_fallible(curve)
```

### Postcondition

**Theorem 1.1.** `composability` returns `Ok(out)` if the composition of a vector of privacy parameters `d_mids` is bounded above by `self.compose(d_mids)` under `adaptivity` adaptivity and `out`-composability. Otherwise returns an error.

*Proof.* The new curve constructed on line 8 composes all epsilon parameters at a given fixed `alpha`. By the postcondition of `InfAdd` we have that for any choice of `alpha`, $\sum_i \texttt{d_mid}_i(\texttt{alpha}) \leq \texttt{compose(d_mids)}(\alpha)$.

[1] See new changes with `git diff f5bb719..8959e35b rust/src/combinators/sequential_composition/mod.rs`

| Adaptivity | Sequential | Concurrent |
|---|---|---|
| Non-Adaptive | Proposition 1[Mir17] | Theorem 2[Lyu22] |
| Adaptive | - | - |
| Fully-Adaptive | Theorem 4.3[FZ22] | Theorem 1.22[VW21] |

This table is reflected in the implementation of `composability` on line 2.

□

# References


[FZ22] Vitaly Feldman and Tijana Zrnic. Individual privacy accounting via a renyi filter, 2022.

[Lyu22] Xin Lyu. Composition theorems for interactive differential privacy, 2022.

[Mir17] Ilya Mironov. Rényi differential privacy. In *2017 IEEE 30th Computer Security Foundations Symposium (CSF)*, page 263–275. IEEE, August 2017.

[VW21] Salil Vadhan and Tianhao Wang. Concurrent composition of differential privacy, 2021.